

SMART SOS DEVICE

That Raises Alarm And Sends Location

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This project aims to develop a wearable safety and alarming device that can be operated easily, especially by children and senior citizens when in distress. It may come handy as the world is constantly witnessing increase in the number of criminal activities. Senior citizens living alone are vulnerable to panic situations due to safety reasons or, sometimes, a medical condition. This device can send an SOS call and simultaneously raise an alarm, if needed.

The device most often used for such a purpose is the SOS button. It is a discreet device that lets the user send an SOS alert to quickly communicate for help from a trusted person by using GSM, GPS, or GPRS. It is used especially for anti-aggression alarms in jewellery stores, banks, or small businesses, and for personal safety.

In this project, we aim to build a customised, cost-efficient version of the SOS button, which is stealthy and uses the best in class GSM-GPS

modules along with a high-decibel buzzer. Its hardware interfacing on breadboard is shown in Fig. 1.

The components used in the project and their function is described below.

SIM800L is a miniature cellular module that allows GPRS transmission (sending and receiving), SMS, and voice calls. It is a low-cost device with small footprint and quad band frequency. The module is a perfect solution for any project that requires long-range connectivity.

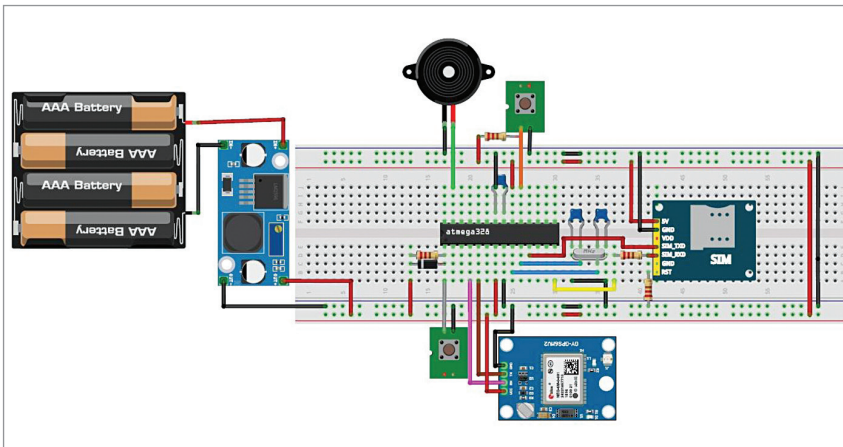


Fig. 1: Hardware interfacing on breadboard

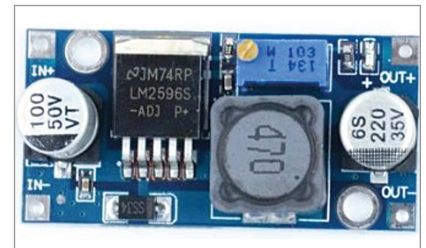


Fig. 3: Buck converter

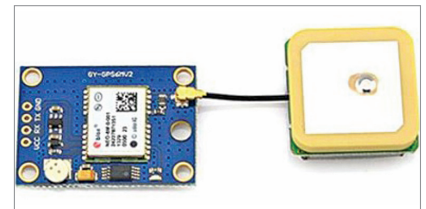


Fig. 4: Neo-6M GPS module



Fig. 2: Pin details of SIM800L

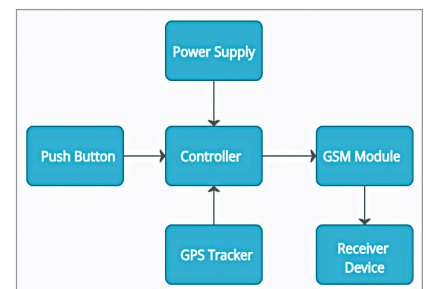


Fig. 5: Block diagram